Best Practices for AutoML's Feature Selection Stage

The main concept behind AutoML's feature engineering approach is expansion-reduction. The goal is to explore as many possible features as it is computationally feasible. Then, only the most relevant features should be chosen.

In the reduction step, AutoML trains a Gradient Boosted Trees (GBT) model on a sample of your development data. The GBT model is used as a surrogate to get its model feature importance. By default this step selects the top-K features so that their relative importance is 90% of the total.

On the advanced options of the AutoML wizard you can select the sample size that will be used for the GBT model.

Feature importance

This methodology is different than the one used by Pulse for individual model importance. Speed is paramount for AutoML. Thus, it's preferable to use GBT on a sample of the training data. This enables faster iterations.

Pulse's feature importance is done by computing the difference in Recall after shuffling each feature individually, for a given FP-Rate. This makes this algorithm model agnostic. However, it also makes it computationally heavy when the number of features is large and/or you have a very unbalanced data set (and thus need a larger sample).

AutoML's feature importance for feature selection is a simpler algorithm that trains a GBT model and gets its feature importance. Only one model is trained, and thus it only goes through the randomly sampled data once.

If your target rate is very low on your features data (e. g. 1 basis point), bear in mind you should tune your feature importance job to use approximately 1M instances.

At the end of the feature selection stage, Pulse will create a new features plan with **only the selected features and their dependencies**.